

Phase 1 Report: Project Proposal

Solar-Powered Off-Grid Charging System for a 48 V Light EV for Emergency Power

Team # 29

A Report
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Table of Contents

Abstract.....	3
Objectives, Requirements, and Specifications	3
System architecture	4
Design information	4
Power converter and component selection	4
MPPT, charging control, and software	5
Measurements and protection.....	5
Emergency power output	5
Physical construction and laboratory use.....	5
Verification and documentation	6
Design Specifications.....	6
Numerical Analysis.....	6
Electrical ratings and operating range.....	6
Measurement of Success	9
Numerical Analysis.....	9
Test Plan.....	10
Review of Existing Solutions	11
Alternatives	12
Key parameters	12
Alternatives selection	12
Project Planning and Schedule	15
Team Formation.....	16
Nadir Chetouani (electrical engineering)	16
Marijan Miskovski (electrical engineering)	17
Mohamed Ali Al-Difai (electrical engineering)	17
Ivan Arapović (electrical engineering).....	17
Gabriel Lauture (computer engineering)	17
Kenny Boum-Biha (computer engineering).....	17
Tasks Breakdown.....	17
Description and Allocation for Each Team Member	17
Communication plan.....	19
Communication Tools.....	19

Discord Organization	20
Meeting Sprint Structure	20
Workflow Tracking with Jira	20
Repository Management with GitHub	20
<i>Contingency Plan</i>	21
Technical Risks	21
<i>Tools Required</i>	22
<i>Budget Estimation</i>	22
<i>References</i>	23

Abstract

The phase 1 report presents a preliminary proposal for a solar powered off-grid charging system for the 48 V battery of a light electric three-wheeler. The system charges the battery entirely from photovoltaic (PV) energy, allowing operation without grid access. Stored energy can also supply low power emergency loads such as phones. The proposed design uses a DC-DC converter with maximum power point tracking and charging control on a TI C2000 F28379D microcontroller. Alternatives were compared. The design will be simulated in MATLAB before a hardware prototype is built and tested. This report covers the preliminary specifications, test plan, team organization, risks and budget.

Objectives, Requirements, and Specifications

The main objective of this project is to design, build and demonstrate a solar-powered charging system. The system will supply energy for the 48 V battery of a light electric three-wheeler. The system will charge the battery without a grid connection. It will also allow stored battery energy to supply selected low-power emergency loads. For example, it will be able to charge phones and lighting. The project addresses charging needs where grid power or charging stations are unavailable. It also explores the use of the vehicle battery as an emergency power source. [9]

The charging performance will mostly depend on the available solar energy. [1] The emergency operating duration will depend on the connected loads.

The project proposal establishes the main deliverables. They are as follow:

- Custom charging converter.
- Microcontroller-based maximum power point tracking (MPPT).
- Justification regarding converter topology.
- Working prototype with operating waveforms. **[9]**

The following requirements define the proposed system. It also defines the information needed to complete its design.

System architecture

The charging system will consist of a photovoltaic (PV) array, input protection, a controlled DC–DC converter and the vehicle battery interface. **[2] [10]** There will be a separate protected output. It will convert battery power to the voltage required by the emergency loads. The other components are the controller, sensors and gate drivers. They must receive power from the PV source. This needs to happen during the normal operation of the system. Whether the system needs to support charging and emergency power delivery at the same time remains to be confirmed with our supervisor.

Design information

The team will collect and verify the information needed to define the design. In this case, this includes PV characteristics, battery charging limits and emergency load needs. **[10]** It also includes project constraints, such as cost, equipment and the project schedule. **[8, Secs. 3.1.1 and 3.2.2]** The PV characteristics will include, in particular, the module model, the rated power, the open-circuit voltage, the short-circuit current and intended series or parallel arrangement. The team will also need to establish its current–voltage and power–voltage curves. The design needs to account for changes in sunlight and temperature. **[1]** This includes the highest expected PV open-circuit voltage. The battery characteristics will include, in particular, the chemistry, the capacity, the operating voltage range, the maximum charging current, the permitted charging temperatures and the charge termination conditions. While the 48 V rating is a good starting point, the actual charging settings must be followed when the information of the battery is determined with the supervisor. It is important that unconfirmed values remain clearly identified until they are established.

Power converter and component selection

The team will design and construct the main DC–DC converter. This includes selecting all the components. **[10]** The components are as follow: power semiconductors, inductor, capacitors, gate drivers, sensors and protection devices. **[9]** The converter topology will be selected using the measured PV characteristics. It will also be selected using the verified battery voltage range. **[9] [4]** Suitable alternatives will be discussed in the “Alternatives” section. The converter must provide controlled charging throughout its operating range.

Component selection will account for voltage, current, temperature and switching stresses. **[4]** The design will also address electrical connections and switching noise. These could affect measurements or control. **[2]** The final numerical specifications will define the converter's characteristics. This includes its continuous power rating, voltage and current ranges, as well as the switching frequency. These ratings must be verifiable.

MPPT, charging control, and software

The proposed controller is the TI C2000 F28379D platform. **[10]** This controller is proposed in the project proposal. **[9]** It will acquire measurements. It will then execute MPPT and charging control. Finally, it will generate switching commands and respond to faults. MPPT will use PV voltage and current measurements to adjust the source operating point as solar conditions change. The algorithm will be determined during the development of the control system. Perturb and Observe and Incremental Conductance are interesting methods that the team found while researching. They could be used as a starting point. **[1]** Battery charging limits needs to take priority over maximum solar power extraction. The controller must reduce charging power or stop charging when the battery reaches a voltage, current, or temperature limit. **[2]** The software will define the various states. The states are as follow: initialization, standby, charging, charge completion and fault. Abnormal situations should not cause uncontrolled charging.

Measurements and protection

The system will measure PV voltage, PV current, battery voltage, and battery charging current. **[2]** Protection will address several situations. For example, overcurrent, overvoltage, overheating, unintended reverse current, incorrect connection polarity, and excessive battery discharge. Depending on the fault, the control software will limit current or disable switching. Critical faults will use hardware protection. The control software will be used for more manageable faults. The system should be able to indicate faults.

Emergency power output

The battery will use a DC–DC stage to supply the selected devices. Low-voltage devices must not be connected directly to the battery. The output must remain within the connected devices permitted supply range. **[4]** Overload and short-circuit protection needs to be integrated to limit the effect of load faults. The project proposal uses DC powered loads.

Physical construction and laboratory use

The prototype will use secure mounting and suitable wiring. Cables, connectors and fuses need to be rated for their intended operating conditions.

Verification and documentation

The team will develop a simulation model of the PV source, converter, battery interface, and controller in MATLAB. The simulations need to support topology selection. It will also need to support the control software control before the construction. The project proposal indicates that the final demonstration will include the demonstration of the system characteristic. For example, the PV voltage, the PV current, the switching-node voltage, the inductor current and battery charging current. [9] Testing will also be used to evaluate the performance of the system. For example, the conversion efficiency, the MPPT performance, the charging, the thermal behaviour and emergency power delivery. The team will maintain clear design records. These will include verified source information, schematics, component selections, simulation files, and test results. [8, Secs. 3.1.3 and 3.2.7]

Design Specifications

Table 1 presents the main electrical and performance specifications. The values taken from the project proposal are fixed requirements. Other numerical values are proposed targets or values from an example component. These remain subject to supervisor review. For preliminary PV specifications, the proposed reference is one Canadian Solar CS6.1-54TM-450H module. Its listed values apply under the manufacturer's standard test conditions. The final operating limits must take into considerations the actual modules, their arrangement, and environmental conditions. TBD means that information is still required, it will be acquired through both tests and datasheets if applicable. "—" means that no separate value is specified. "Typ." indicates a nominal value or a proposed design setting.

Numerical Analysis

Electrical ratings and operating range

Table 1: Electrical Ratings and Operating Range						
#	Parameter	Min.	Typ.	Max.	Unit	Comments
1.	Nominal battery voltage	—	48	—	V	Battery voltage class specified in the project proposal [9]. The charger must match this class. It is not a value to maximize.
2.	Battery operating voltage range	TBD	—	TBD	V	Lowest and highest permitted voltages. No reliable range can be given without the battery chemistry and model. Matching this range supports correct converter selection.
3.	Charging voltage setpoint	—	TBD	—	V	Voltage maintained during voltage-regulated charging. It must follow the battery's

						instructions. Nominal 48 V does not establish it. Correct selection prevents excessive charging voltage.
4.	Maximum charging current	—	—	TBD	A	Highest permitted charging current. More current can shorten charging time. However, a value requires the battery/BMS limits and converter ratings.
5.	Battery energy capacity	—	TBD	—	Wh	Stored energy available for vehicle and emergency use. Greater usable capacity supports longer operation. The battery's capacity is not known
6.	PV rated power	—	450	—	W	Proposed module rating from the reference panel [12] .
7.	PV voltage at maximum power	TBD	33.0	TBD	V	Voltage where the example panel produces its rated maximum power [12] . Matching the converter to this voltage improves energy extraction. Environmental limits remain to be established.
8.	PV open-circuit voltage	—	38.9	TBD	V	Panel voltage with no load [12] . Input components must tolerate the highest expected value. This requires the minimum expected temperature and final array configuration.
9.	PV input current	—	13.66	TBD	A	Example current at maximum power [12] . Adequate current capacity limits overheating and losses. The maximum must also account for short-circuit current and expected sunlight.
10.	Continuous converter output power	—	400	—	W	Proposed charging-power rating. The information comes from TI's 400 W design [2] . Higher power can shorten charging time but increases component and cooling demands. This is subject to change.
11.	Converter switching frequency	—	100	—	kHz	Starting value used by the boost stage in TI's Solar Explorer example [10] . Selection balances smaller

						filtering components against switching losses.
12.	DC–DC conversion efficiency	95	—	—	%	Share of input power delivered to the battery. Higher efficiency reduces losses and heating. Proposed below Renogy’s advertised conversion performance of over 97% [13] .
13.	MPPT tracking efficiency	98	—	—	%	Share of available maximum PV power extracted. Higher is better. Proposed slightly below Renogy’s advertised tracking performance of over 99% [13] .
14.	Battery voltage regulation error	—	—	2	%	Maximum absolute difference from the voltage command. Lower error improves charging precision. Proposed as a less demanding target than the BQ25756’s listed $\pm 0.5\%$ regulation [14] .
15.	Battery current regulation error	—	—	5	%	Maximum absolute difference from the current command. Lower error improves current control. Proposed as a less demanding target than the BQ25756’s listed $\pm 3\%$ regulation [14] .
16.	Emergency output voltage	TBD	12	TBD	V	Proposed intermediate DC output. This follows Microchip’s example [11] . It supports compatible lighting and a suitable phone-charging adapter. The output voltage must remain within the operating voltage requirements of the connected devices. This is subject to change
17.	Emergency output rated power	—	24	—	W	Proposed from Microchip’s 12 V, 2 A output: $12 \times 2 = 24 \text{ W}$ [11] . More capacity supports larger loads. However, their consumption determines battery runtime. This is subject to change

Conversion efficiency will be evaluated as follow:

$$\eta_{\text{conv}} = \frac{P_{\text{battery}}}{P_{\text{PV,input}}} \times 100\%$$

MPPT tracking efficiency will be evaluated as follow:

$$\eta_{\text{MPPT}} = \frac{P_{\text{PV,operating}}}{P_{\text{PV,maximum}}} \times 100\%$$

MPPT efficiency describes how closely the controller operates near the available maximum power. Conversion efficiency describes the losses between the PV input and battery output.

Measurement of Success

The success of the proposed project will be evaluated on its ability to efficiently and safely convert photovoltaic energy into controlled charging power for a 48V battery, all while providing the emergency low-voltage power.

Numerical Analysis

The system's performance will be assessed using measurable parameters and compared against the target values:

Table 2: Measurement of Success Criteria	
Parameter	Target value
Nominal battery voltage	48 V ± manufacturer specification
Continuous charging power	400 W
DC-DC converter efficiency	≥ 95%
MPPT tracking efficiency	≥ 98%
Battery voltage regulation error	≤ 2%
Battery current regulation error	≤ 5%
Emergency load	12 V
Emergency power	24 W

5 key areas will be investigated for the success criteria:

1. Solar energy harvesting and MPPT performance, the efficiency will be determined by comparing the extracted PV power with the maximum available solar power under the same operating conditions.
2. DC-DC converter performance will be evaluated by comparing the input and output power.

3. Battery charging accuracy will be assessed through voltage and current measurements, ensuring that the charging process remains within the battery's limits. The protection mechanisms will also be verified under abnormal operating conditions.
4. Emergency power delivery will be tried by ensuring the system can provide a regulated and consistent supply of power.
5. Overall system integration and off-grid functionality will be tested by harvesting solar energy and operating without any connection to the grid.

Test Plan

The proposed system will undergo simulation and experimental testing to verify its performance against the established design specifications. Initial simulations will be conducted in MATLAB to evaluate the converter topology and control strategy before hardware implementation. Laboratory testing will be performed on each individual subsystem and finally a complete systems test.

1. **Solar energy harvesting and MPPT performance:**
The MPPT controller will be tested under different levels of sunlight, a solar simulator would be ideal, but such equipment is very costly, we will resort to a solar array simulator. The PV voltage and current will be recorded to calculate the extracted power. These measurements will be compared with the maximum available power obtained from the PV power-voltage curve. The controller should achieve a minimum tracking efficiency of 98% under conditions where the battery can accept power.
2. **DC-DC Converter Performance:**
The converter will be tested under different input voltages and loading conditions. Input and output voltage and current will be measured using laboratory instruments to calculate the conversion efficiency, of which the minimum is 95%. It also needs to supply 400W continuously when sufficient power is available. An oscilloscope will be used to observe switching waveforms and evaluate output stability.
3. **Battery charging and protection:**
The charging process will be monitored to verify that the controller maintains a maximum voltage regulation error of 2% and current regulation error of 5%. Measurements will be compared against the reference charging values established for the selected 48V battery. Additionally, protection mechanisms will be evaluated through controlled fault simulations, including overvoltage, overcurrent and overheating to verify that the controller appropriately limits or interrupts charging.
4. **Emergency Power Delivery:**
The emergency output will be tested using a DC electronic load or compatible low-power appliances. The output voltage and current will be measured under different

loading conditions to verify that the system can provide a regulated 12VDC output at 24W. Output stability and overload protection will also be evaluated.

5. Overall System Integration and Off-Grid Functionality:

Following individual subsystem testing, the complete prototype will be evaluated without an external grid connection. The test will demonstrate energy harvesting from the PV source, controlled charging of the 48V battery, and emergency power delivery using the stored battery energy. Measurements will be taken throughout the system to verify proper interaction between the converter, MPPT, controller, battery and emergency output.

All results will be compared with the MATLAB simulation results, and the measurement of success criteria will determine whether the project objectives have been achieved.

Review of Existing Solutions

In the current market, solar-powered battery systems are already commercially available in several forms. These systems range from simple MPPT charge controllers to integrated solar inverter/charger systems. The fundamental architecture in those systems is also proposed in this project: using photovoltaic energy to charge a battery through a controlled power converter. However, most available solutions are designed as general-purpose energy products rather than specifically for the off-grid charging of a 48 V light electric vehicle combined with low-power emergency energy delivery.

The Victron Energy SmartSolar MPPT family of solar charge controllers is a good example of existing solutions. These devices take power generated by a photovoltaic array and convert it into controlled charging power while using MPPT to maximize the energy extracted from the PV source. Many models support 48 V battery banks. For example, the SmartSolar MPPT RS is designed for 48 V battery systems and supports PV input voltages that range from 65 V to 450 V. This device also provides programmable battery charging characteristics. **[20]**

Another example of an existing solution is the Victron EasySolar-II GX. This device combines an MPPT solar charge controller, inverter/charger and a system controller within one enclosure. The difference between this device and the SmartSolar MPPT RS is that the EasySolar-II GX can both store photovoltaic energy in a battery and also use the stored energy to supply electrical loads. This architecture performs similar functions to both the charging and emergency-power portions of the proposed project. **[21]**

Although these commercial products provide similar functionality, the proposed system is intended for a more specific application. The design will be focused on the 48 V battery of a light electric three-wheeler. It will use a custom DC-DC converter which will be controlled by a microcontroller-based MPPT algorithm. A separate protected output will allow energy

stored in the vehicle battery to supply selected low-power emergency loads. Unlike commercially integrated solutions, the converter, sensing, control and protection stages will be design and experimentally evaluated as part of the project. This will allow a better optimization of the converter topology and operating parameters depending on the selected PV array, vehicle battery and target charging power rather than using a general-purpose commercial charger.

Alternatives

Key parameters

The alternatives considered for the charging converter will be evaluated according to a set of key technical and practical parameters. Since the system is powered by a photovoltaic source, conversion efficiency is an important parameter because any converter loss directly reduces the amount of solar energy available for battery charging. The converter also needs to provide a suitable voltage conversion range which needs to be between the PV array and the 48 V vehicle battery. The topology that is selected must be compatible with the microcontroller-based MPPT control since the PV operating point will be adjusted through the converter duty cycle. The circuit complexity and the component count are also considered as key parameters because the converter must be designed, constructed and tested within the available project time. Semiconductor voltage and current stresses are evaluated because they affect component selection, reliability and cost. The voltage ripple is considered due to the effect on battery charging quality. Finally, the last key parameter is the cost which is a constraint when selecting different components. **[16] [17] [18]**

Here is the efficiency equation:

$$\eta = \frac{P_{out}}{P_{in}} * 100\%$$

And the voltage ripple equation:

$$Voltage\ ripple = \frac{\Delta V_o}{V_o} * 100\%$$

Alternatives selection

Three DC-DC converter topologies will be selected as possible solutions for the PV battery charging stage: the buck converter, boost converter, and buck-boost converter. These three alternatives all represent the main possible relationships between the photovoltaic array voltage and the required battery charging voltage. The buck converter is used if the PV array voltage remains above the required battery voltage meaning it requires a step-down conversion. The boost converter is used if the PV voltage remains below the battery charging voltage and needs to be increased. Finally, the buck-boost can be used to both step-up and step-down operation which allows the system to operate over a wider PV voltage range. **[15][16]** Each of these topologies will be evaluated based on efficiency, voltage conversion capability, MPPT controllability, circuit complexity, component stress, output ripple and cost.

Table 3: Alternatives comparison				
Parameter	Importance (%)	Buck	Boost	Buck-Boost
Conversion efficiency	20	9	8	7
Voltage conversion capability	20	6	6	10
MPPT controllability	15	8	9	8
Circuit simplicity / Component count	15	10	9	6
Component voltage/current stress	10	9	7	6
Output ripple performance	10	9	7	6
Cost	10	10	9	6
Total importance	100	8.50	7.80	7.30

For the table above, 10 is the most favorable and 1 the least favorable.

The total importance of each alternative was calculated using this equation:

$$Total\ Importance = \sum \left(\frac{I_i}{100} * R_i \right)$$

I_i = importance of parameter i

R_i = alternative rating from 1 – 10

The importance percentages were chosen based on the priorities of the project. Efficiency and voltage conversion capability had the highest importance because the system must make good use of the limited solar energy and must be able to interface correctly between the PV array and the 48 V battery. MPPT controllability and circuit simplicity were given a relatively high importance because MPPT is required in the system and the converter must be realistic to design, build and test. Component stress, output ripple and cost were given a lower but still significant importance because they affect reliability, charging quality and the practicality of the final prototype.

The conversion efficiency scores were chosen based on the expected losses of each converter topology. The buck converter received the highest score because of its simple

power stage, generally low conduction and low switching losses. The boost converter was rated slightly lower because the switch and the inductor carry the input current continuously which can create more electrical stress on the components. The buck-boost converter received the lowest score because it requires a more complex power path and also requires additional switching devices which increase switching losses. [15]

For the voltage conversion capability, the buck and boost converters received moderate scores because each one can only operate in one direction. As mentioned previously, a buck converter can only step-down the voltage and a boost converter can only step-up the voltage. The buck-boost converter received the highest score because it can both increase and decrease the voltage which makes it more flexible when the PV voltage varies. [15] [16]

For MPPT controllability, all three converters received relatively high scores because their duty cycle can be adjusted by a microcontroller to control the operating point of the PV array. The control of buck converter and boost converter are common in PV applications because their control is relatively simple. [17] The wider operating range and more complex topology of the buck-boost converter can make the control more difficult. [18]

For circuit simplicity, the buck converter received the highest score because it has a simple structure with a low number of power components. The boost converter is also simple and therefore received a similar score. The buck-boost converter was rated lower because it requires more switches, more gate-driving circuits and more complicated control.

For component voltage and current stress, the buck converter received the best score because the voltage and current stress on the component are generally easier to manage. The boost converter was rated lower because the switch and diode must be able to resist the boosted output voltage. The buck-boost converter received the lowest score because some implementations can expose the switches to a higher combined voltage and current stresses which makes component selection more difficult.

For output ripple, the buck converter received the highest score because its output inductor naturally smooths the current delivered to the battery. This is useful for battery charging because the charging current is less pulsating. The boost converter received a lower score because the output current goes through a diode in a more pulsating way which makes the output capacitor more important. [16] The buck-boost converter was rated slightly lower because depending on the topology the output current can be more pulsating than the boost converter. [19]

For the cost, the buck converter received the best score because it can be implemented with a small number of components and simple control circuit. The boost converter received a similar score because it also requires a low number of components. The buck-

boost converter received the lowest score because it requires more switches, gate drivers and control circuits which increases the expected cost of the converter.

Project Planning and Schedule

The project will be completed progressively throughout the different phases of the capstone project. The main work is divided into research, simulation and design, hardware development, firmware development, testing and final system integration. Some of these tasks will be done at the same time since the project includes both hardware and software work.

At the beginning of the project, the main focus will be on defining the system requirements and doing the research needed for the design. This includes comparing the different converter topologies, determining the characteristics of the PV source and battery, selecting the main components and defining the general system architecture. Preliminary simulations will also be started in MATLAB.

Once the main design parameters are confirmed, the team will focus more on the detailed design of the DC-DC converter, sensing circuits, protection circuits and control system. Firmware development for the TI C2000 F28379D will also begin during this stage. This includes reading voltage and current measurements using the ADC, generating PWM signals, controlling the converter and eventually implementing the MPPT algorithm.

After the design has been verified through simulation, the project will move toward the hardware stage. The required components will be ordered, and the different circuits will be built and assembled. The power stage, sensing circuits and other hardware sections will first be tested individually before being connected together.

The last part of the project will mainly focus on integrating the complete system and testing its performance. The converter, MPPT controller, battery charging system and emergency output will be tested together. The results obtained from the prototype will be compared with the simulation results and with the performance targets defined earlier in the report. Time will also be kept near the end of the project for troubleshooting, adjustments, the final demonstration and the final report.

The team will work using three-week sprints. Tasks will be assigned and followed using Jira, and the progress of each member will be discussed during the weekly team meetings. When possible, tasks such as firmware development, simulation and component research will be worked on in parallel.

Table 4: Monthly activity schedule	
Period	Main Activities

September – Early October	Team formation, project definition, initial research, system requirements and architecture
October 2026	Converter topology analysis, PV and battery research, preliminary component selection and Phase 1 proposal
November 2026	Detailed converter calculations, MATLAB modelling, sensing and protection design, initial MCU familiarization
December 2026	Converter simulation validation, preliminary control implementation, component selection and preparation for hardware development
January 2027	Hardware design, PCB/schematic development, component ordering, PWM and ADC firmware development
February 2027	Convert assembly and subsystem testing, closed-loop control development and initial MPPT implementation
March 2027	MPPT testing, battery charging tests, protection testing and emergency output implementation
Late March – Early April 2027	Complete system integration, off-grid testing, efficiency measurements and comparison with simulation
April 2027	Troubleshooting, final improvements, final demonstration, documentation and final report

Team Formation

The team is made up of six students from Electrical and Computer Engineering. Nadir Chetouani, Marijan Miskovski, Mohamed Ali Al-Difai and Ivan Arapović are Electrical Engineering students, while Gabriel Lauture and Kenny Boum-Biha are Computer Engineering students. This works well for the project because it includes both electrical and software/control work. The Electrical Engineering students will be more involved in the power electronics, converter design, circuit design and electrical measurements. The Computer Engineering students will focus more on the microcontroller, firmware, control implementation and hardware/software integration. Some parts of the project will require both sides to work together. For example, the sensing circuits need to be designed electrically, but the signals also need to be read and processed correctly by the microcontroller. The MPPT and control sections will also require both simulation and firmware implementation. The team will still work together on testing, integration, documentation and the final demonstration.

Nadir Chetouani (electrical engineering)

Nadir Chetouani has experience in software development and power engineering. His skills will support the design and development of the DC–DC converter prototype and the microcontroller software.

Marijan Miskovski (electrical engineering)

With his very good understanding and experience with control systems and electronics, Marijan will be a key asset to design the circuit and the logic that will be used within it. Marijan has and is currently taking graduate level courses in which he has completed numerous design projects for control systems, power electronics and linear systems, which will bring important value to the project's design.

Mohamed Ali Al-Difai (electrical engineering)

Mohamed brings a mix of electrical and programming skills that will help connect the hardware and software sides of the project. He can help with simulation, data analysis and supporting the firmware team when needed. His experience in data analysis and visualization will be useful for processing test results and comparing measured performance with simulations.

Ivan Arapović (electrical engineering)

With his experience in electrical tests in locomotive overhaul such as continuity, insulation resistance, high potential tests, etc. Ivan will take lead for the testing portion of the project. He also has hands-on industry experience in picking electrical components and building electrical circuits, which will be a key part of the project.

Gabriel Lauture (computer engineering)

Gabriel Lauture is a Computer Engineering student with experience in programming, embedded systems and hardware/software integration. His main contribution to the project will be on the TI C2000 microcontroller, firmware development, ADC measurements, PWM generation and the implementation of the MPPT and control algorithms.

Kenny Boum-Biha (computer engineering)

Kenny Boum-Biha is a Computer Engineering student with experience in programming, MATLAB Simulink, hardware/software integration and control systems. Currently working as a software developer intern for OPAL-RT Technologies, he provides useful insight in simulation and power electronics.

Tasks Breakdown

Description and Allocation for Each Team Member

The project is divided into several main tasks based on the different parts of the system. Since many parts of the project are connected, more than one team member can be assigned to the same task. The responsibilities are also divided based on the background

of each member, with the Electrical Engineering students focusing more on the power electronics and hardware sections, while the Computer Engineering students focus more on the microcontroller, firmware and control implementation.

Table 5: Task Breakdown		
Task	Team Members	Main Work
System requirements and research	All members	Define the system requirements, research the PV source, battery, converter options and required components.
PV and battery characterization	Nadir, Marijan, Mohamed	Determine the PV operating characteristics, battery voltage range, charging limits and other electrical requirements.
DC-DC converter design	Nadir, Marijan, Ivan	Perform converter calculations, select the power components and determine the final converter parameters.
MATLAB simulation	Nadir, Ivan, Mohamed	Develop and test the converter and system models before hardware implementation.
Sensing and protection	Mohamed, Marijan, Gabriel	Design the voltage and current sensing circuits, protection circuits and make sure the signals can be properly read by the microcontroller.
TI C2000 setup and firmware	Gabriel, Kenny	Configure the F28379D, ADC inputs, PWM generation, system states and other embedded firmware required for the system.
MPPT and control	Gabriel, Kenny, Ivan	Develop, simulate and implement the MPPT algorithm and the converter control logic.
Hardware assembly and integration	Nadir, Marijan, Mohamed, Ivan, Gabriel, Kenny	Assemble the different hardware sections and

		connect the power stage, sensing circuits and controller together.
Emergency power output	Ivan, Mohamed, Kenny	Develop and test the low-voltage emergency output and its control and protection.
Testing and validation	All members	Test each subsystem, measure the performance of the complete system and compare the experimental results with the simulations and project specifications.
Documentation and final report	All members	Document the design process, simulation results, hardware results and testing results throughout the project.

Communication plan

The team consists of six members who work on hardware, firmware and simulation tasks all simultaneously. This communication plan keeps everyone aligned while avoiding work. The team uses slightly altered Agile methodology: three-week sprints, a weekly team meeting rather than a daily and a small number of dedicated tools.

Communication Tools

Table 6: Communication Tools		
Tool	Purpose	Users
Discord	Day-to-day messaging, quick questions, voice calls for remote work sessions	All members
Jira	Sprint planning, task tickets, workload tracking, progress visibility	All members
GitHub	Version control and code review for MCU firmware (MPPT, control loops), simulation scripts and models	Software/control members (access for all)

Concordia email	Formal communication with the supervisor, Technical Coordinator and Academic Coordinator	Team leader, plus CC to all members
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Discord Organization

The team server is split into topic channels so that discussions remain searchable and do not drown each other which is necessary due to the long duration of the project. Channels include #power-stage, #control-firmware, #simulation, #pcb-hardware, #reports and #meetings. When Discord decisions affect scope, specifications or schedule they are summarized, posted and pinned in the #meetings channel. This will then be copied into the related Jira ticket so that no important information stays only in chat history. The team members are expected to reply to mentions within twenty-four hours on weekdays.

Meeting Sprint Structure

The team meets once a week for one hour either in person at the Capstone Laboratory or on a Discord voice call. In every meeting each member answers three questions: what was worked on in the last week, what will be worked on this week and is anything blocking progress. Meeting times will be tracked using the monthly log.

Each three-week sprint starts with a planning meeting separate than the weekly meeting on a predetermined day. In that meeting, tickets are drawn from the Jira backlog and assigned to members according to each members role in the task list. The sprint ends with a combined review and retrospective. During the review, completed work is demonstrated and adjusted if need be. Sprint deadlines will be set to match course deadlines in order to ensure a sprint review always happens at one week before each report submission, before the Final Design Approval and before the in-lab demos. This schedule gives time for integration and final editing.

Workflow Tracking with Jira

All work is tracked as Jira tickets that belong to epics matching the project's subsystems: PV characterization, converter power stage, magnetics and passive components MPPT and control firmware, protection and sensing PCB design, testing and documentation. Each ticket has an owner, a story-point estimate reflecting effort and a due date set for the sprint. Tickets move through the backlog with set these statuses: To Do, In Progress, In Review and Done. The backlog provides the team and supervisor with a quick view of workload balance. Additionally, the backlog history supplies evidence for the Phase 2 schedule update and for individual contribution reports.

Repository Management with GitHub

A GitHub repository will be created for the purpose of harboring firmware for the TI C2000 F28379D, MATLAB/Simulink models and PCB design files. When working on the repository,

team members are expected to develop on feature branches named after their associated Jira ticket (for example: MPPT-12-incremental-conductance). Completed changes are then merged into the branch only via a pull request that is reviewed by at least one other member. This keeps the branch in a working state and spreads code knowledge across the team. Large binary files, such as Simulink models are committed with messages so that changes can be traced.

Contingency Plan

Since this project necessitates a large amount of customization and tuning (e.g. power hardware, embedded firmware, etc.), risk levels are inevitable and need to be accounted for. To counter this and ensure smoothness of procedure, it is imperative to have contingency plans.

Each risk below is rated for likelihood (L) and impact (I) as Low, Medium or High. For each one, we list a preventive measure that lowers the chance it happens and a contingency action that we apply if it happens anyway. Each determined risk will be mentioned in its concerned Jira ticket and will have a special label called *risk*. Risks will be reviewed at every sprint retrospective.

Technical Risks

Table 7: Technical Risks				
Risk	L	I	Prevention	Contingency
Measured PV curves do not suit the chosen topology (for example, PV voltage at MPP falls inside or below the battery voltage window)	Medium	High	Characterize the PV modules early in Phase 2, before finalizing the topology	Reconfigure the series/parallel arrangement of the PV modules, or switch to a buck-boost stage. Simulations are kept modular so the change can be made quickly.
Synchronous buck gate-drive problems (shoot-through, dead-time tuning)	Medium	Medium	Simulate dead time in PSIM/Simulink and choose a driver with built-in dead-time protection	Fall back to asynchronous operation by disabling the low-side switch and using a freewheeling diode, accepting lower efficiency at high current as reported in the literature

MPPT algorithm fails to converge or oscillates on hardware	Medium	Medium	Validate the algorithm in simulation and on the PV emulator before outdoor tests	Use a simpler, well-proven algorithm (Perturb & Observe with a fixed step), or fixed-duty or constant-voltage operation, to demonstrate charging
Insufficient sunlight for the Phase 4 outdoor test (Montréal winter, cloud cover, snow)	High	Medium	Plan several possible outdoor test dates in March and record any sunny-day data as early as possible	Demonstrate using the PV emulator with measured I–V curves, and present the recorded outdoor data as supporting evidence

Tools Required

The tools required for this projected are separated in 4 sections: test equipment, software, hardware and project-management tools. Most of the test equipment and software are expected to be available through the Capstone Laboratory.

The test equipment consists of; an oscilloscope with differential voltage and current probes that will capture the waveforms required: PV voltage and current, switching-node voltage, inductor current and battery charging current. A DC power supply or a PV simulator will be necessary to simulate repeatable PV curves, so the MPPT algorithm can be tested indoors not matter the type of environment and weather. Multimeter and soldering tools will be used for general measurements, assembly and repairs.

The software needed for this project includes MATLAB/Simulink with Simscape Electrical and PSIM. These will be used to model the PV source, converter, battery and controller before building the hardware. Code Composer Studio will be used to develop and debug the firmware on the F28379D.

Lastly, the team will use project management tools such as discord for communication, Jira for task tracking and GitHub for version control.

Budget Estimation

Table: Estimated Prototype Budget

Table 8: Budget estimation			
#	Item	Average Unit cost \$	Total \$
1	PV module, Solar panel	250	250

2	TI LAUNCHXL-F28379D LaunchPad	65	65
3	48 V battery pack with BMS (e.g., LiFePO ₄ , ~20 Ah)	275	275
4	Power MOSFETs	6	6
5	Inductor / magnetics (core and winding or off-the-shelf)	—	25
6	Capacitors	—	30
7	Protection (fuses, breaker, TVS, reverse-polarity protection)	—	35
8	PCB fabrication	300	300
9	Enclosure	15	15
10	Miscellaneous ICs and passive components, wiring, connectors	—	60
	Hardware subtotal		1061
11	Oscilloscope, probes, PV emulator, electronic load, multimeters, soldering station	Lab-provided	0
12	MATLAB/Simulink, Simscape Electrical, Embedded Coder, PSIM, Code Composer Studio	University license / Free	0
13	Discord, Jira, GitHub	Free plans	0
	Contingency (15%)		~159
	Estimated total		~1220

The budget is an estimate based on the team's own judgement of what a prototype would need. Since the final design and component has not yet been decided, the listed items, quantities and prices are not final and may change. The budget will be revised once the final bill of material has been completed.

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